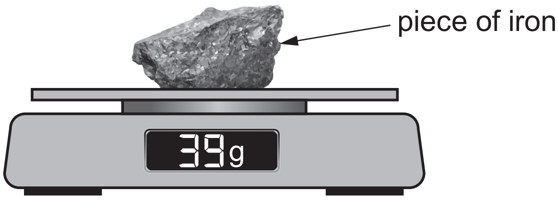
3. Peter carries out an experiment to find the density of iron.

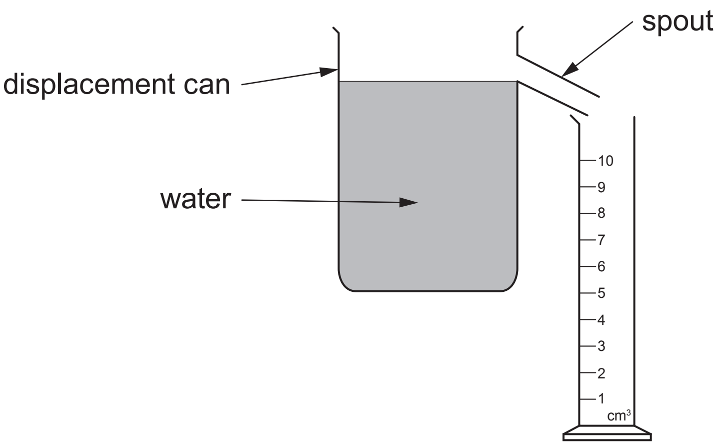
Stage 1

- He uses a balance to measure the mass of a piece of iron.



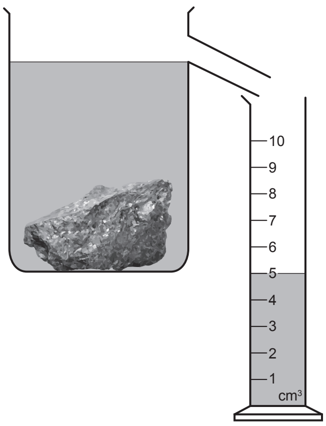
Stage 2

- He fills a displacement can with water.
- He waits until water stops flowing from the spout.
- He then places an empty measuring cylinder under the spout.



Stage 3

- He carefully lowers the iron into the water.
- He collects the water displaced in the measuring cylinder.



Examiner only

Use information from the diagrams to answer the following questions.

(i) State the mass of the iron. mass = ... g [1]

(ii) State the volume of the iron. volume = ... cm^3 [1]

(iii) I. Use the equation:

$$\text{density} = \frac{\text{mass}}{\text{volume}}$$

to calculate the density of iron. [2]

density = ...

II. Underline the correct unit that should be used for the density above. [1]

g/cm^3 **g/cm^2** **cm^2/g** **cm^3/g**

(iv) State **one** change to the apparatus that would improve the accuracy of the measurement of the **mass** of the iron. [1]

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